LOOK INSIDE

PKG-002 · Water & Sanitation Pack

Sample pages from: Rainwater Harvesting Purification Plumbing Layout

These are unaltered pages from the actual guide, taken from the current fixed library (Stage 4, 2026-10-30). Every EOTWSG guide opens with an action card, states its safety boundaries up front, and ends with printable fields for your household.

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Rainwater Harvesting & Purification Plumbing Layout

EOTWSG | MODULE 03 | REV 2026-09-22 | STATUS: SOURCE-VERIFIED ONLY

Authored by: EOTWSG

REVISION LOG

Date	Revision	Description
2026-09-22	REV 2026-09-22	Front-matter standardization: cover block, revision log, action card, and local fields added; source normalized to house Typst dialect.

ACTION CARD

Use this guide to credit: Excerpts, schematics, and tactical protocols adapted from the End of the World Survival Guide (EOTWSG) by Matthew Danielson. Reproduce only for off-grid resilience and survival training.. Carry one copy in the master box and one copy in the working binder.

Step	Do this now	Completion check	Stop / escalate when
1	Confirm this copy is current: check the module ID (03) and REV date on the cover.	Module ID and REV noted on the working copy.	If the REV predates 2026-09-22, replace this copy before use.
2	Work through the Sizing Basis section against your household's actual people, conditions, and equipment.	Section completed with household-specific entries; gaps flagged.	If a required capability is missing, flag it in the binder and resolve before the next drill.
3	Record completion, date, and any deviations in the working binder; schedule the next review.	Binder entry dated and filed.	If the household situation changes, re-run the guide.

PURPOSE AND ACTIVATION

Purpose: Credit: Excerpts, schematics, and tactical protocols adapted from the End of the World Survival Guide (EOTWSG) by Matthew Danielson. Reproduce only for off-grid resilience and survival training.

Activation triggers: utility failure, evacuation order, sudden service uncertainty, or a planned verification drill. This module is usable on its own; linked module IDs preserve the shared decision order from the master index.

600 sq ft Catchment — 40 in Rainfall — Multi-Stage Treatment

Sizing Basis

Input	Value	Review note
Catchment area	600 sq ft roof footprint	Confirm actual contributing area, bypasses, overflow, and roof material.
Annual rainfall	40 in / 1,000 mm	Use local rainfall records and design storm intensity for storage and overflow.
Efficiency factor	0.95 metal-roof factor	Confirm first-flush, debris, losses, and actual runoff conditions.

Resolved Yield Formula

ANNUAL USABLE YIELD

Catchment Area (sq. ft.) × Rainfall (inches) × 0.623 × 0.95 = Usable Yield (gallons)

600 sq ft × 40 inches × 0.623 × 0.95 = 14,204 gallons/year

Exact Filtration Sequence

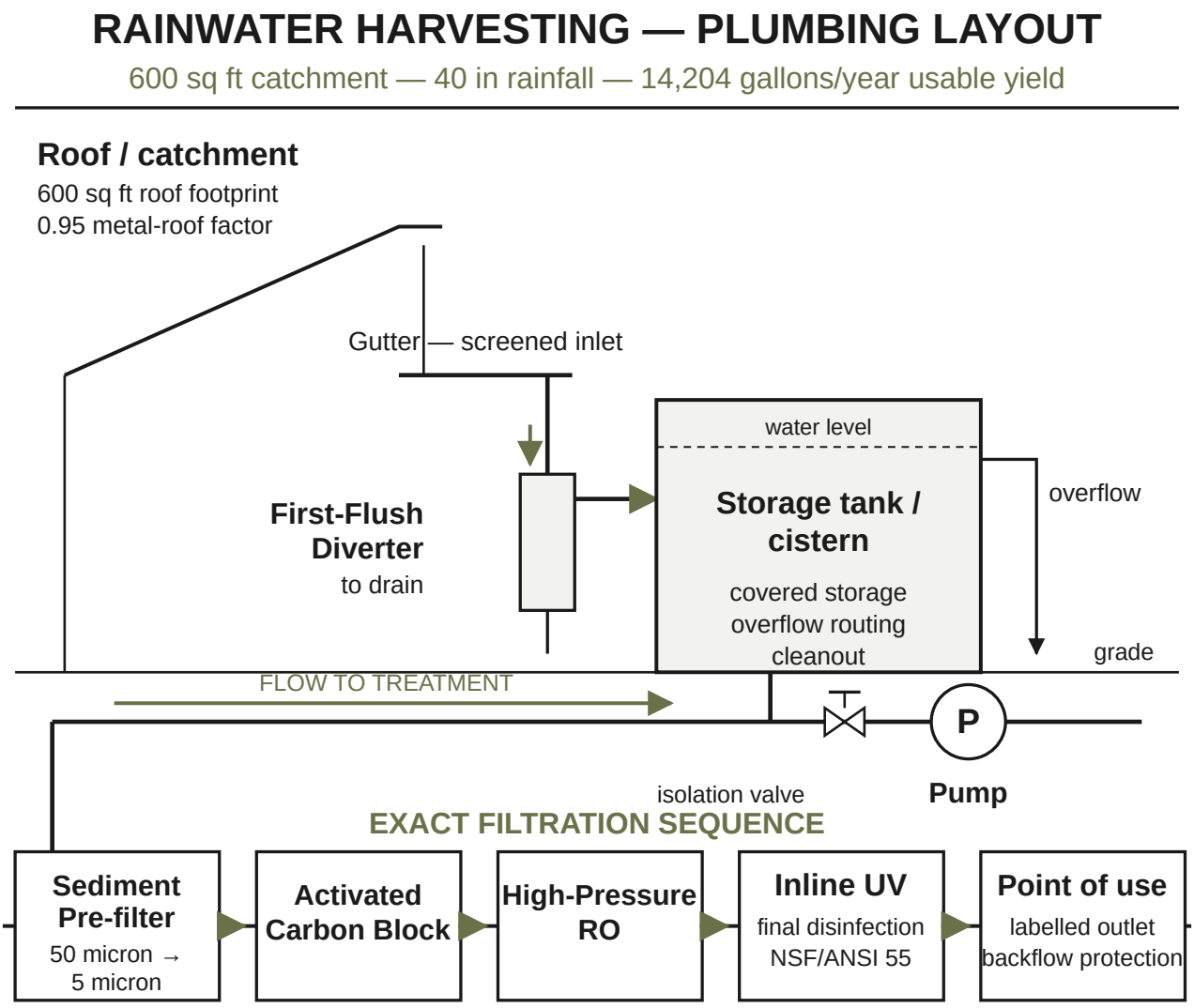


Figure 1: Rainwater harvesting and purification plumbing layout: the exact filtration sequence.

FINAL DISINFECTION

Route treated water through an inline UV disinfection lamp after the high-pressure RO stage. Specify components evaluated for NSF/ANSI Standard 53 (filtration of cysts/organics) and Standard 55 (UV microbiological disinfection), and verify the exact product listing, flow rate, dose, maintenance, and monitoring requirements.

Plumbing and Safety Review

1. Provide screened roof inlets, covered storage, overflow routing, cleanouts, isolation valves, drain points, and backflow protection as required by the AHJ.
2. Treat harvested rainwater as non-potable until the complete system is engineered, commissioned, tested, and approved for the intended use.
3. Maintain filter change records, UV lamp hours, turbidity or quality checks, storage sanitation records, and a bypass lockout for failed treatment stages.
4. Separate potable and non-potable piping, label all outlets, and prevent cross-connections.
5. Size storage, pumps, RO reject flow, pressure controls, and emergency overflow for the local site and actual demand.

WATER SAFETY

Untreated or inadequately treated rainwater can contain pathogens, chemicals, and particulates. Do not drink it based on appearance or taste. A professional must verify treatment performance, cross-connection control, plumbing code, and public-health requirements.

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SAFETY BOUNDARIES

WARNING - MODULE LIMITS

Do not perform medical, electrical, or structural procedures beyond this guide's stated limits. When in doubt, stop and consult qualified professionals. This guide is a preparedness aid, not a guarantee of safety. Follow current instructions from local emergency management and authorities.

- This module directs specific actions; those actions carry the safety limits stated above. Do not improvise beyond them.
- Use nonviolent de-escalation and withdrawal as the primary security response.
- If conditions exceed the household's capability, escalate to professional responders.

LOCAL FIELDS AND RECORDS

Use these printable fields and keep them at the front of the working binder:

Field	Fill-in
Household / location name	**Name:** _____
Primary contact	**Name and number:** _____
Date completed	**Date:** _____
Next review date	**Date:** _____